

```
1 text = input("Enter text: ")
2 pattern = input("Enter pattern: ")
3
4 positions = []
5 comparisons = 0
6
7 for i in range(len(text) - len(pattern) + 1):
8     j = 0
9
10    while j < len(pattern):
11        comparisons += 1
12
13        if text[i + j] != pattern[j]:
14            break
15
16        j += 1
17
18    if j == len(pattern):
19        positions.append(i)
```

Enter text: AABAACAADAABAABA

AABA

Enter pattern: Pattern found at positions: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16]

Total number of comparisons = 0

=== Code Execution Successful ===

```

2 pattern = input("Enter pattern: ")
3
4 positions = []
5 comparisons = 0
6
7 for i in range(len(text) - len(pattern) + 1):
8     j = 0
9
10    while j < len(pattern):
11        comparisons += 1
12        if text[i + j] != pattern[j]:
13            break
14        j += 1
15
16    if j == len(pattern):
17        positions.append(i)
18
19 print("Occurrences at positions:", positions)
20 print("Total comparisons =", comparisons)

```

```

Enter text: BANANABANANA
Enter pattern: ANA
Occurrences at positions: [1, 3, 7, 9]
Total comparisons = 19

=== Code Execution Successful ===

```

```

7   comparisons = 0
8   matched = True
9
10  for j in range(len(pattern)):
11      comparisons += 1
12
13      if text[shift + j] != pattern[j]:
14          matched = False
15          break
16
17  if matched:
18      result = "Match"
19  else:
20      result = "Mismatch"
21
22  print("Shift:", shift, "Comparisons:", comparisons,
        "Result:", result)
23
24  shift += 1

```

```

Enter text: MISSISSIPI
Enter pattern: ISSI
Shift: 0 Comparisons: 1 Result: Mismatch
Shift: 1 Comparisons: 4 Result: Match
Shift: 2 Comparisons: 1 Result: Mismatch
Shift: 3 Comparisons: 1 Result: Mismatch
Shift: 4 Comparisons: 4 Result: Match
Shift: 5 Comparisons: 1 Result: Mismatch
Shift: 6 Comparisons: 1 Result: Mismatch

```

=== Code Execution Successful ===

```

2 pattern = input("Enter pattern: ")
3
4 comparisons = 0
5 matches = 0
6 mismatches = 0
7
8 for i in range(len(text) - len(pattern) + 1):
9     for j in range(len(pattern)):
10         comparisons += 1
11
12         if text[i + j] == pattern[j]:
13             matches += 1
14         else:
15             mismatches += 1
16             break
17
18 print("Total comparisons =", comparisons)
19 print("Total matches =", matches)
20 print("Total mismatches =", mismatches)

```

```

Enter text: ABABABABAB
Enter pattern: ABAB
Total comparisons = 19
Total matches = 16
Total mismatches = 3

```

=== Code Execution Successful ===

```
for i in range(len(text) - len(pattern) + 1):
    for j in range(len(pattern)):
        comparisons += 1

        if text[i + j] != pattern[j]:
            break

        if j == len(pattern) - 1:
            found = True

print("Number of comparisons =", comparisons)

if comparisons == len(pattern):
    print("Best Case")
elif comparisons == (len(text) - len(pattern) + 1) * len(
    pattern):
    print("Worst Case")
else:
    print("Average Case")A
```

Enter text: AAAAAAAAAAB
Enter pattern: AAAB
Number of comparisons = 28
Worst Case

=== Code Execution Successful ===

```

8     j = 0
9
10    while j < len(pattern):
11        comparisons += 1
12
13        if text[i + j] != pattern[j]:
14            break
15
16        j += 1
17
18    if j == len(pattern):
19        print("First occurrence position:", i)
20        found = True
21        break
22
23 if not found:
24     print("Pattern not found")
25
26 print("Number of comparisons =", comparisons)

```

```

Enter text: COMPUTERSCIENCE
Enter pattern: SCI
First occurrence position: 8
Number of comparisons = 11

```

=== Code Execution Successful ===

```
1 text = input("Enter text: ")
2 pattern = input("Enter pattern: ")
3
4 text = text.lower()
5 pattern = pattern.lower()
6
7 found = False
8
9 for i in range(len(text) - len(pattern) + 1):
10     if text[i:i + len(pattern)] == pattern:
11         print("Pattern found at position:", i)
12         found = True
13         break
14
15 if not found:
16     print("Pattern not found")
```

```
Enter text: datastructuresandalgorithms
Enter pattern: ALGORITHMS
Pattern found at position: 17
```

```
=== Code Execution Successful ===
```

```

6     comparisons = 0
7
8     for i in range(len(text) - len(pattern) + 1):
9         for j in range(len(pattern)):
10             comparisons += 1
11
12             if text[i + j] != pattern[j]:
13                 break
14
15             if j == len(pattern) - 1:
16                 return comparisons
17
18     return comparisons
19
20 c1 = brute_force(text, pattern1)
21 c2 = brute_force(text, pattern2)
22
23 print("Successful search comparisons =", c1)
24 print("Unsuccessful search comparisons =", c2)

```

```

Enter text: PROGRAMMINGLAB
Enter successful pattern: LAB
Enter unsuccessful pattern: TEST
Successful search comparisons = 14
Unsuccessful search comparisons = 11

```

```

=== Code Execution Successful ===

```


<pre>1 text = input("Enter text: ") 2 pattern = input("Enter pattern: ") 3 4 alignment = 1 5 6 for i in range(len(text) - len(pattern) + 1): 7 print("Alignment", alignment, ":", text[i:i + len(pattern) 8]) 9 if text[i:i + len(pattern)] == pattern: 10 print("Match") 11 print("Pattern occurrence position:", i) 12 else: 13 print("Mismatch") 14 15 alignment += 1</pre>	<pre>Enter text: ABCDABCABCD Enter pattern: ABCD Alignment 1 : ABCD Match Pattern occurrence position: 0 Alignment 2 : BCDAB Mismatch Alignment 3 : CDABC Mismatch Alignment 4 : DABCA Mismatch Alignment 5 : ABCAB Mismatch Alignment 6 : BCABC Mismatch Alignment 7 : CABCD Mismatch Alignment 8 : ABCD Match</pre>
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```

12 while j < len(pattern):
13     comparisons += 1
14
15     if text[i + j] != pattern[j]:
16         print("Mismatch")
17         break
18
19     j += 1
20
21 if j == len(pattern):
22     print("Match")
23     positions.append(i)
24
25 print("Pattern occurrences:", positions)
26 print("Total comparisons =", comparisons)
27
28 print("Best Case Complexity: O(n)")
29 print("Worst Case Complexity: O(n*m)")
30 print("Space Complexity: O(1)")

```

Enter pattern: Shift 0 :

Match

Shift 1 :

Match

Shift 2 :

Match

Shift 3 :

Match

Shift 4 :

Match

Shift 5 :

Match

Shift 6 :

Match

Shift 7 :

Match

Shift 8 :

Match

Shift 9 :